Teaching Plan of AAE1001 (2026/2027 Semester 1)

1. Subject Title and Subject Code
Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering (AAE1001)

2. Enrolment and Class Size
84 Students (48403 -)

3. Subject Intended Learning Outcome (ILO)

Upon completion of the subject, students will be able to:

(i) Demonstrate an understanding of the foundational concepts of Artificial Intelligence and Data Analytics (AIDA);

(ii) Acquire basic skills in using AIDA technologies and applications;

(iii) Articulate examples of how the adoption AIDA could enhance their understanding on aeronautical and aviation engineering; and

(iv) Demonstrate an awareness of global contemporary ethical issues and impact from AIDA applications in daily life.

4. Grading Policy:

(a) Weighting of this course: 100% Continuous assessment

(b) Continuous assessment (CA)

e-Learning module	**(15%)**
Assignment	**(25%)**
Laboratory	**(35%)**
Group project presentation	**(25%)**

*** The e-Learning module is a compulsory component of this subject. Incompletion of the e-Learning module will lead to failure in this subject.**

TEACHING PLAN

AAE1001 – Introduction to Artificial Intelligence and Data Analytics in Aerospace and Aviation Engineering
Plan of Teaching, Learning and Assessment –Semester I, 2026/2027

Instructor:
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Time & Venue:
Lecture: Mon 16:30-18:20
Venue: Y304

Week	Topics Taught	Planned Learning Outcomes	Assessment	Timetable, Venue, Staff
Wk 1 (Aug 31)	Overview of AI and Data Analytics in Aerospace and Aviation Engineering	• History and concepts of AI. • Familiar with basic concepts of AIDA and how relevant technologies are applied in Aerospace and Aviation Engineering.	e-Learning Module (compulsory, completed by **Oct 31**)	**2 hours** LEC/TUT Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 2 (Sep 7)	Fundamentals of Data Analytics *Career Development Talk by SAO (5:50 pm)*	• Understand basic concepts in data analytics. • Familiar with the basic formats and types of data. • Conduct basic data analysis and recognize data-quality problems		**2 hours** LEC/TUT Mon 16:30-18:20 Y304 Dr. Guohao ZHANG

Wk 3 (Sep 14)	Fundamentals of Machine Learning	• Understand basic concepts and theories of machine learning. • Familiar with linear regression and classification in supervised learning and its applications.		**2 hours** LEC/TUT Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 4 (Sep 21)	Model Evaluation and Extension to Deep Learning	• Distinguish model performance on training data from generalization to unseen data • Identify underfitting and overfitting from simple results or visualizations • Understand basic concepts of deep learning.	Assignment 1 (submit by **Oct 19** via Blackboard)	**2 hours** LEC/TUT Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 5 (Sep 28)	Apply AI and Data Analytics in Aerospace and Aviation Engineering	• How AI is used for Self-piloted Airplanes, Aircraft Maintenance, Aerospace Manufacturing, Air Traffic Management, and Airport management. • AI in satellite and aerospace engineering.		**2 hours** LEC/TUT Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 6 (Oct 5)	Group Project Overview Introduction of GitHub	• Understanding the framework of engineering project using AIDA • Understanding the importance of GitHub • Installation & environment setup for a GitHub-based project • Understanding the basic functions of GitHub	Assignment 2 (submit by **Nov 2** via Blackboard)	**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 7 (Oct 12)	Introduction on Path Planning Group Project Task 1-3	• Understanding the principle of path planning • Able to implement path planning on aviation tasks by codes		**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG

Wk 8 (Oct 19)	No Lecture - *General holiday (The day following Chung Yeung Festival)*			
Wk 9 (Oct 26)	Collaborative Coding using GitHub	• Understanding the operation of GitHub in a teamwork • Understanding the basic components of a coding project • Able to conduct collaborative coding using GitHub for the group project		**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 10 (Nov 2)	Group Project Task 4-6	• Able to create new components in the template code to solve advanced tasks • Able to implement AI for the aviation tasks in the group project		**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 11 (Nov 9)	Group Project Conclusion	• Understanding the role of collaborative coding and AIDA in aeronautical and aviation engineering		**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 12 (Nov 16)	Group Project Presentation I	• Technical report and presentation skills • Project management and collaboration on engineering tasks	Presentation of the group project results for each group (all members are required to present)	**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG
Wk 13 (Nov 23)	Group Project Presentation II	• Technical report and presentation skills • Project management and collaboration on engineering tasks	• Presentation report • GitHub repository • Peer assessment evaluation (submit by **Nov 28**)	**2 hours** LEC/LAB Mon 16:30-18:20 Y304 Dr. Guohao ZHANG

Remarks: N/A